

COMP6203 INTELLIGENT AGENTS
UTILITY AND GAME THEORY WORKSHEET

1. QUESTION 1

Let $\Omega = \{\omega_1, \dots, \omega_8\}$ be a set of outcomes. Suppose that an agent has the following preference ordering over Ω :

$$\omega_1 \succ \omega_2 \succeq \omega_3 \succeq \omega_4 \succ \omega_5 \succeq \omega_6 \succeq \omega_7 \succeq \omega_8.$$

Which of the following utility functions $u : \Omega \rightarrow \mathbb{R}$ represent the above preference?

- (1) $u(x) = \frac{1}{i}$, for $x = \omega_i$
- (2) $u(x) = \prod_{j=1}^i \frac{1}{j}$, for $x = \omega_i$
- (3) $u(x) = \begin{cases} 7 & \text{if } x = \omega_1, \\ 5 & \text{if } x = \omega_i, 2 \leq i \leq 4 \\ 3 & \text{if } x = \omega_i, 5 \leq i \leq 8 \end{cases}$
- (4) $u(x) = \begin{cases} 3 & \text{if } x = \omega_1 \\ 5 & \text{if } x = \omega_i, 2 \leq i \leq 4 \\ 7 & \text{if } x = \omega_i, 5 \leq i \leq 8 \end{cases}$
- (5) $u(x) = 10$ for all $x \in \Omega$.
- (6) $u(x) = \sum_{j=1}^i \frac{1}{j}$, for $x = \omega_i$
- (7) $u(x) = \prod_{j=1}^i (-j)$, for $x = \omega_i$
- (8) $u(x) = \sum_{j=1}^i (-j)$, for $x = \omega_i$

Solution. Any function representing the above preference ordering must be non-increasing. Only (1), (2), (3) and (8) represent the ordering correctly.

2. QUESTION 2

Consider the following strategic-form game

		Player 2				
		F	G	H	I	J
Player 1	A	5 7	4 1	4 4	3 7	3 4
	B	4 4	5 3	5 8	8 1	3 4
	C	1 6	3 3	3 5	2 6	1 4
	D	4 3	6 1	8 2	10 3	2 3
	E	5 3	5 1	4 2	8 3	3 3

- (1) Which of the above strategies are weakly dominated (by some other strategy)?
- (2) Compute the pure Nash equilibria of the game after eliminating weakly dominated strategies.
- (3) Compute the whole set of pure Nash equilibria.

Solution.

- (1) A is weakly dominated by E. C is strictly (and therefore weakly) dominated by B. G is strictly (and therefore weakly) dominated by F. I is weakly dominated by F. J is weakly dominated by F.
- (2) You can choose which strategy to eliminate first. Depending on your choice, the process of elimination will likely be different. Every time a strategy is eliminated, the game changes and gets smaller and we need to look for weakly dominated strategies in the smaller game. The image below shows the result of eliminating strategies A, G, C, and I (in that order). The equilibria of the resulting game (obtained by looking for best responses) are (E, F) and (E, J) .

		Player 2				
		F	G	H	I	J
Player 1	A					
	B	4 4		5 8		3 4
	C					
	D	4 3		8 2		2 3
	E	5 3		4 2		3 3

- (3) Pure strategy Nash equilibria of the whole game are $(A, F), (D, I), (E, F), (E, J)$ as shown in the following image.

		Player 2				
		F	G	H	I	J
Player 1	A	5 7	4 1	4 4	3 7	3 4
	B	4 4	5 3	5 8	8 1	3 4
	C	1 6	3 3	3 5	2 6	1 4
	D	4 3	6 1	8 2	10 3	2 3
	E	5 3	5 1	4 2	8 3	3 3

3. QUESTION 3

Consider the following game. One night both your house and your neighbour's get burglarized. The burglar steals exactly the same items from both houses. You and your neighbour file a claim with the same insurance company asking for compensation for the burglary. The insurance company contacts you and your neighbour separately and asks each of you to estimate the value of what was stolen. You can claim a maximum of £1000 and a minimum of £50, but how much money you receive depends on how honest you are with your estimation. If both you and your neighbour ask for the same amount of money, you both get exactly that amount. If you ask for different amounts, you will both receive the smaller amount with a penalty/reward of £50: so, if you ask for x and y with $x < y$, the person asking for x will receive $\mathcal{L}(x + 50)$, while the person asking for y will receive $\mathcal{L}(x - 50)$. Suppose that at no point you can communicate with your neighbour and that each of you can claim any amount of money from

$$M = \{\mathcal{L}50, \mathcal{L}60, \mathcal{L}70, \dots, \mathcal{L}980, \mathcal{L}990, \mathcal{L}1000\}.$$

You can see this as a strategic-form game where you and your neighbours are the players.

What is the set of all pure strategy Nash equilibria?

Solution. This is a game similar to the traveler's dilemma. If both players choose £1000, they both get £1000 but one of them (say player 1) benefits from deviating to £990, which increases their payoff but decreases the payoff for player 2. Player 2 then is incentivised to match player 1 and play £990, but, again, player 1 can deviate by asking for a lower amount, etc. This shows that the only equilibrium is when players both choose £50.

Formally, take any

$$x, y \in \{\mathcal{L}60, \mathcal{L}70, \dots, \mathcal{L}980, \mathcal{L}990, \mathcal{L}1000\},$$

(where x is player 1's strategy and y is player 2's strategy).

- If $x = y$, then for any $x' = x - \mathcal{L}10$, for player 1, $u_1(x', y) > u_1(x, y)$ (similar for player 2).
- If $x > y$, then for player 1 $u_1(y, y) > u_1(x, y)$. Similar for player 2 if $x < y$.

(£50, £50) is an equilibrium because each player will receive £0 by unilaterally deviating.

4. QUESTION 4

Consider the following strategic-form game with two players P1 and P2. Let $k > 0$ be any natural number. Each player has the same strategy set

$$S = \left\{ 0, \frac{1}{k}, \frac{2}{k}, \dots, \frac{k-1}{k}, 1 \right\}.$$

The following are the utility functions for P1 and P2, respectively:

$$u_1(x_1, x_2) = |x_1 - x_2|, \quad u_2(x_1, x_2) = \sqrt[4]{1 - (|x_1 - x_2|)^4}.$$

Compute the set of equilibria of the game.

Solution. Take any strategy combination (x, y) , where x is a strategy of P1 and y is a strategy of P2. P1 can always try to maximise the distance from P2 by deviating to 0 or 1. On the contrary, P2 can maximise their utility by trying to match P1's choice. So, the set of pure strategy equilibria is $\emptyset$.

Formally, take any

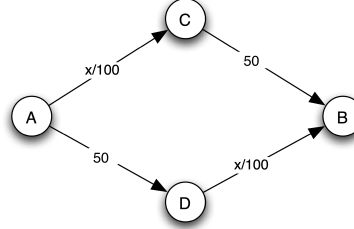
$$x, y \in \left\{ 0, \frac{1}{k}, \frac{2}{k}, \dots, \frac{k-1}{k}, 1 \right\},$$

- If $x = y$, then for any $x' \neq x$, for P1, $u_1(x', y) > u_1(x, y)$.
- If $x \neq y$, then for $y' = x$, $u_2(x, y') > u_2(x, y)$ for P2.

So the game contains no pure strategy Nash equilibria.

5. QUESTION 5

Consider the following road network with four vertices A , B , C , and D :



There are 4000 drivers, each of whom wants to travel from A to B . Each driver chooses a route independently, with the objective of minimising their own travel time.

The travel times on the network's edges are as follows:

- The travel time on $A \rightarrow C$ is $x/100$ minutes, where x is the number of cars using that edge.
- The travel time on $C \rightarrow B$ is 50 minutes.
- The travel time on $A \rightarrow D$ is 50 minutes.
- The travel time on $D \rightarrow B$ is $x/100$ minutes, where x is the number of cars using that edge.

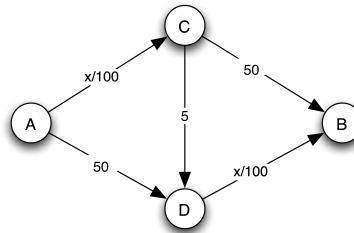
Represent the above situation as a strategic-form game where the drivers are the players of the game and their utility function is defined in terms of the travel time experienced by that player. Compute all the pure-strategy Nash equilibria of the game.

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Consider now a modified version of the above game, obtained by adding the directed edge

$$C \rightarrow D,$$

whose travel time is 5 minutes, independently of the number of cars using it:



Change the formal definition of the previous game to reflect the changes to the network. Compute all the pure-strategy Nash equilibria of the game. How do they compare to the equilibria of the previous game?

Solution. In the first network we have two strategies: ACB , ADB . For each player i , and each strategy $s \in \{ACB, ADB\}$

$$u_i(s) = \begin{cases} -(50 + \frac{x_{AC}}{100}) & \text{if } s = ACB \\ -(50 + \frac{x_{DB}}{100}) & \text{if } s = ADB \end{cases}$$

where x_e is the number of players choosing edge e .

Every outcome where there is a different number of cars between the routes is not an equilibrium, as each car on the more trafficked route can decrease their travel time by deviating to the route with fewer cars. Every outcome where the cars are distributed evenly between the two routes (i.e. 2000 cars on the upper route and 2000 cars on the lower route) is an equilibrium, with payoff -70.

In the second network we have three strategies: ACB , ADB , $ACDB$. For each player i , and each strategy $s \in \{ACB, ADB, ACDB\}$

$$u_i(s) = \begin{cases} -(50 + \frac{x_{AC}}{100}) & s = ACB \\ -(50 + \frac{x_{DB}}{100}) & s = ADB \\ -(5 + \frac{x_{AC}}{100} + \frac{x_{DB}}{100}) & s = ACDB \end{cases}$$

All the situations where players all choose the same route $ACDB$ are equilibria (and the only ones). In those cases though, the payoff of each player is -85, which is worse than the original network where cars were given only two choices.